

Computational Physics

Fourier transforms

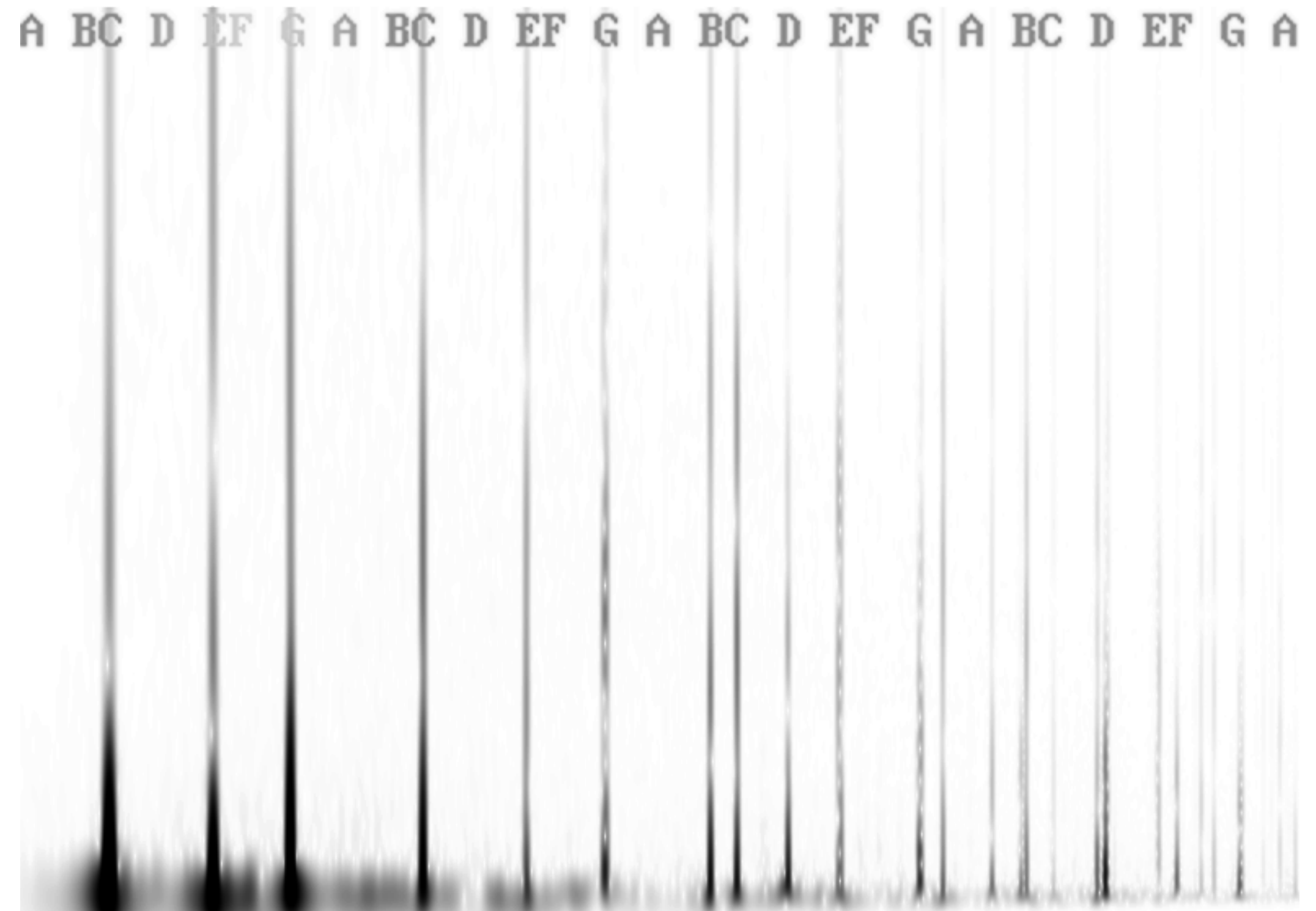
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Fourier transforms

The **Fourier transform** (FT) is **an integral transform that takes a function as input, and outputs another function that describes the extent to which various frequencies are present in the original function.**

The output of the transform is a complex-valued function of frequency. The term Fourier transform refers to both the mathematical operation and to this complex-valued function. When a distinction needs to be made, the output of the operation is sometimes called the frequency domain representation of the original function. The Fourier transform is analogous to decomposing the sound of a musical chord into the intensities of its constituent pitches.



The Fourier transform applied to the waveform of a C major piano chord (with logarithmic horizontal (frequency) axis). The first three peaks on the left correspond to the fundamental frequencies of the chord (C, E, G). The remaining smaller peaks are higher-frequency overtones of the fundamental pitches.

Fourier transforms



The Fourier transform relates the time domain, in red, with a function in the domain of the frequency, in blue. The component frequencies, extended for the whole frequency spectrum, are shown as peaks in the domain of the frequency.

Fourier transforms

Fourier transform

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi\xi x} dx, \quad \forall \xi \in \mathbb{R}. \quad \textbf{(Eq.1)}$$

Inverse transform

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi\xi x} d\xi, \quad \forall x \in \mathbb{R}. \quad \textbf{(Eq.2)}$$

Fourier transforms

In 1822, Fourier claimed that **any function**, whether continuous or discontinuous, **can be expanded into a series of sines**. That important work was corrected and expanded upon by others to provide the foundation for the various forms of the Fourier transform used since.

In general, the coefficients are complex numbers, which have two equivalent forms

$$\hat{f}(\xi) = \underbrace{Ae^{i\theta}}_{\text{polar coordinate form}} = \underbrace{A \cos(\theta) + iA \sin(\theta)}_{\text{rectangular coordinate form}}.$$

$$\begin{aligned}\hat{f}(\xi) \cdot e^{i2\pi\xi x} &= Ae^{i\theta} \cdot e^{i2\pi\xi x} \\ &= \underbrace{Ae^{i(2\pi\xi x + \theta)}}_{\text{polar coordinate form}} \\ &= \underbrace{A \cos(2\pi\xi x + \theta) + iA \sin(2\pi\xi x + \theta)}_{\text{rectangular coordinate form}}.\end{aligned}$$

Fourier transforms

The **discrete-time Fourier transform (DTFT)** is a form of Fourier analysis that is applicable to a **sequence of discrete values**.

The DTFT is often used to analyze samples of a continuous function. The term discrete-time refers to the fact that **the transform operates on discrete data, often samples whose interval has units of time**. From uniformly spaced samples **it produces a function of frequency that is a periodic summation of the continuous Fourier transform of the original continuous function**.

In simpler terms, when you take the DTFT of regularly-spaced samples of a continuous signal, you get **repeating** (and possibly overlapping) **copies of the signal's frequency spectrum**, spaced at intervals corresponding to the sampling frequency.

Fourier transforms

A fast Fourier transform (FFT) is an algorithm that computes the discrete Fourier transform (DFT) of a sequence, or its inverse (IDFT).

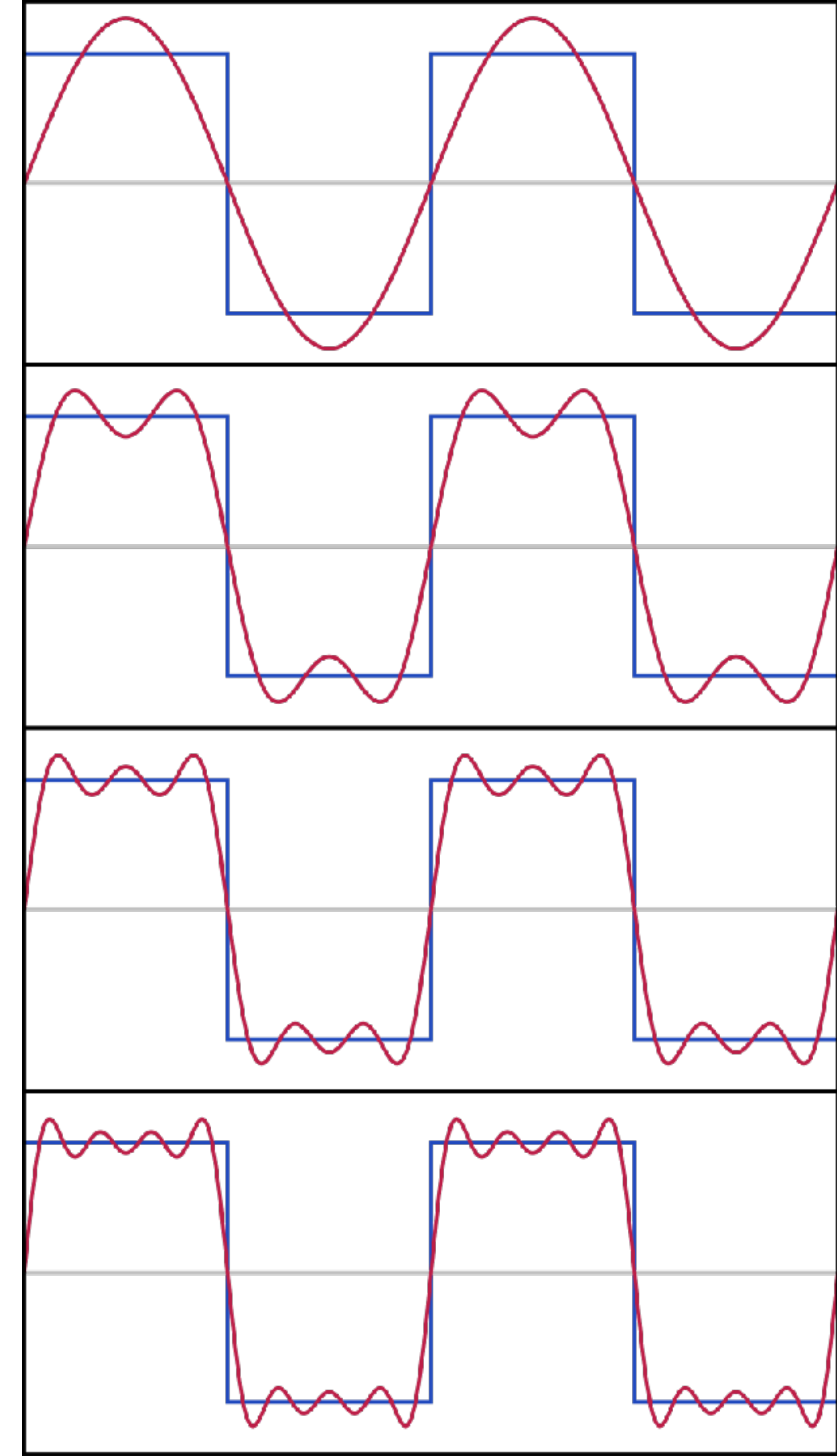
A Fourier transform converts a signal from its original domain (often time or space) to a representation in the frequency domain and vice versa.

Fourier series

A Fourier series is a series expansion of a periodic function into a sum of trigonometric functions. The Fourier series is an example of a trigonometric series. By expressing a function as a **sum of sines and cosines**, many problems involving the function become **easier to analyze** because **trigonometric functions are well understood**.

For example, Fourier series were first used by Joseph Fourier to find solutions to the heat equation. This application is possible because the derivatives of trigonometric functions fall into simple patterns.

Fourier series cannot be used to approximate arbitrary functions, because most functions have infinitely many terms in their Fourier series, and the series do not always converge. **Well-behaved functions**, for example smooth functions, **have Fourier series that converge to the original function**. The coefficients of the Fourier series are determined by integrals of the function multiplied by trigonometric functions, described in Fourier series § Definition.



Fourier series

Sine-cosine form

$$s_N(x) = a_0 + \sum_{n=1}^N \left(a_n \cos\left(2\pi \frac{n}{P} x\right) + b_n \sin\left(2\pi \frac{n}{P} x\right) \right) \quad \textbf{(Eq.1)}$$

Exponential form

$$s_N(x) = \sum_{n=-N}^N c_n e^{i2\pi \frac{n}{P} x} \quad \textbf{(Eq.2)}$$